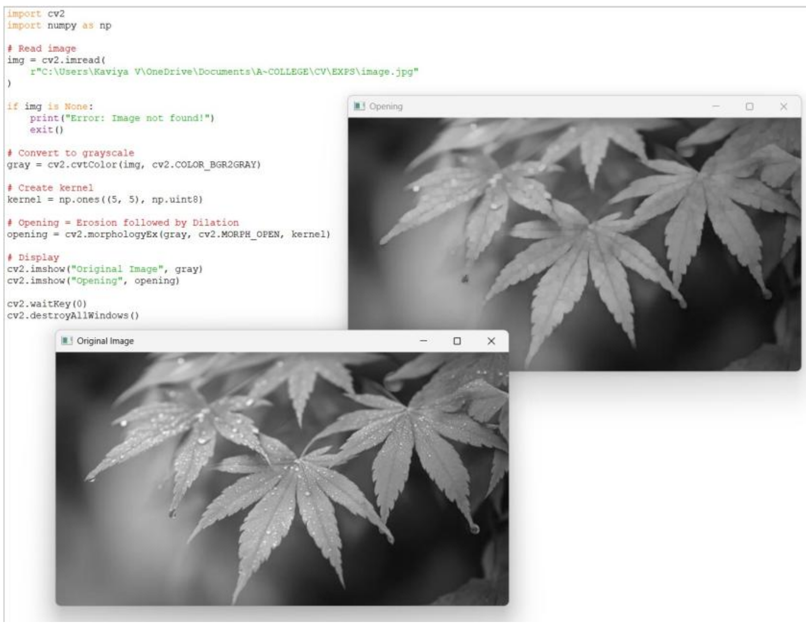
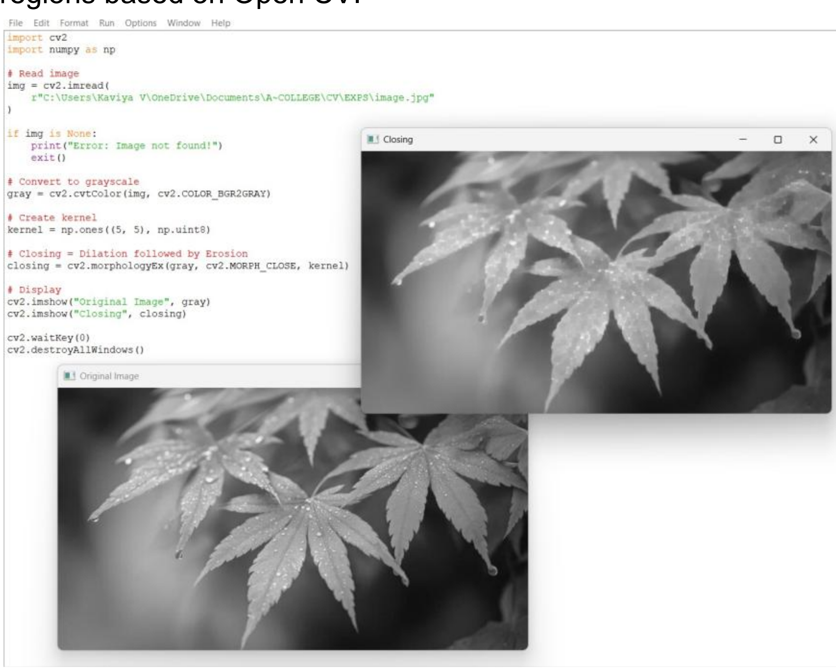
# LAB EXPERIMENTS

**NAME: M.MANASA**
**REG NO: 192425189**
**COURSE CODE: ITA0512**
**COURSE NAME: COMPUTER VISION**

21. Implement the Opening technique as a Morphological operation to dilate the foreground regions based on Open CV.

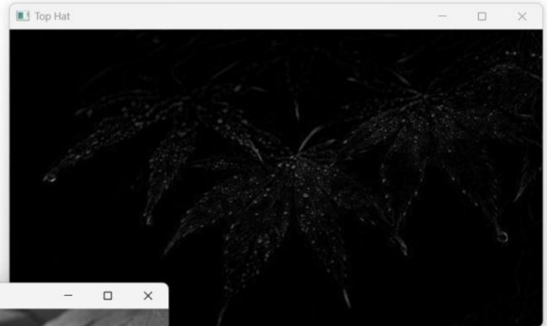


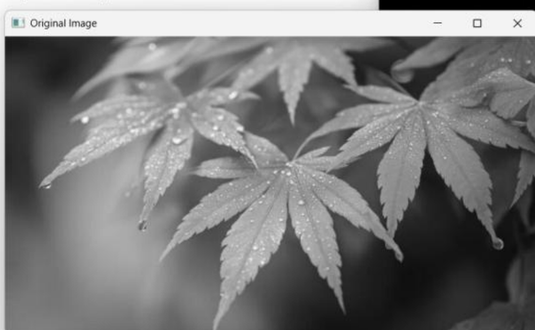

22. Implement the Closing technique as a Morphological operation to dilate the foreground regions based on Open CV.



23. Implement the Top hat technique as a Morphological operation to dilate the foreground regions based on Open CV.

```
import cv2
import numpy as np

# Read image
img = cv2.imread(
    r"C:\Users\Kaviya V\OneDrive\Documents\A-COLLEGE\CV\EXPS\image.jpg"
)

if img is None:
    print("Error: Image not found!")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Top Hat
tophat = cv2.morphologyEx(gray, cv2.MORPH_TOPHAT, kernel)

# Display
cv2.imshow("Original Image", gray)
cv2.imshow("Top Hat", tophat)

cv2.waitKey(0)
cv2.destroyAllWindows()
```




24. Implement the Black hat technique as a Morphological operation to dilate the foreground regions based on Open CV.

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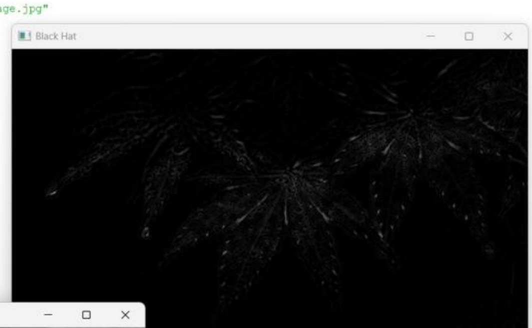
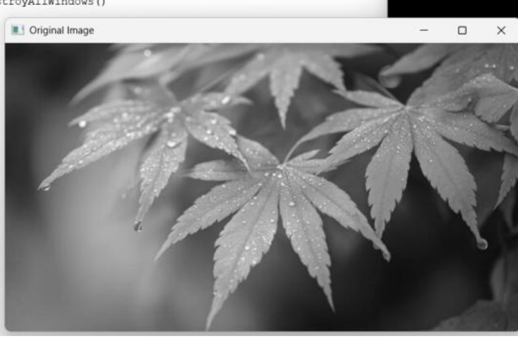
```
import cv2
import numpy as np

# Read image
img = cv2.imread(
    r"C:\Users\Kaviya V\OneDrive\Documents\A-COLLEGE\CV\EXPS\image.jpg"
)

if img is None:
    print("Error: Image not found!")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Black Hat
blackhat = cv2.morphologyEx(gray, cv2.MORPH_BLACKHAT, kernel)

# Display
cv2.imshow("Original Image", gray)
cv2.imshow("Black Hat", blackhat)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



25. Recognize watch from the given image by general Object recognition using Open CV.

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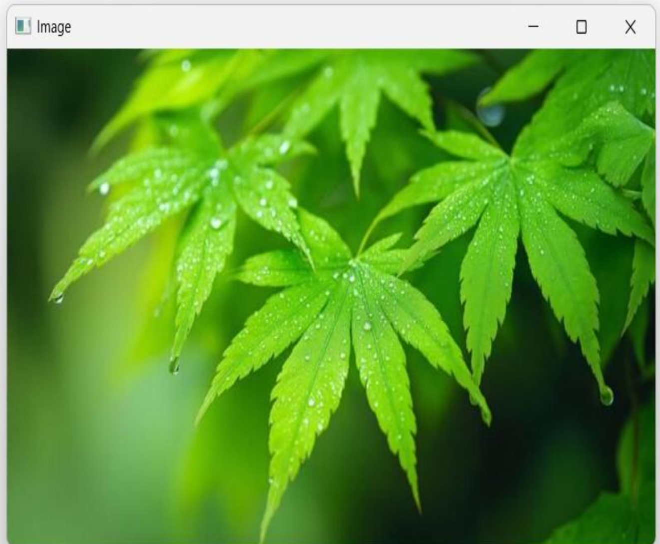
```python
import cv2

img = cv2.imread(
    r"C:\Users\Kaviya V\OneDrive\Documents\A~COLLEGE\CV\EXPS\image.jpg"
)

if img is None:
    print("Error: image.jpg not found!")
    exit()

# Display image
cv2.imshow("Image", img)

print("Select the watch region using your mouse.")
print("Press ENTER after selecting the region.")
print("Press ESC to cancel.")

# Select ROI
x, y, w, h = cv2.selectROI("Image", img, False)

# Crop watch
watch = img[y:y+h, x:x+w]

# Save cropped watch
cv2.imwrite(
    r"C:\Users\Kaviya V\OneDrive\Documents\A~COLLEGE\CV\EXPS\watch.jpg",
    watch
)

cv2.destroyAllWindows()

print("watch.jpg created successfully!")
```




26. Implement a function to reverse the frames of the video to create a video in reverse mode using Open CV.

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```python
import cv2

# Read video
cap = cv2.VideoCapture(
    r"C:\Users\Kaviya V\Downloads\video.webm"
)

if not cap.isOpened():
    print("Error: Video not found!")
    exit()

# Get video properties
fps = cap.get(cv2.CAP_PROP_FPS)
width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))

# Store frames
frames = []

while True:
    ret, frame = cap.read()

    if not ret:
        break

    frames.append(frame)

cap.release()

# Create output video
fourcc = cv2.VideoWriter_fourcc(*"mp4v")

out = cv2.VideoWriter(
    "reverse_video.mp4",
    fourcc,
    fps,
    (width, height)
)

# Write frames in reverse order
for frame in reversed(frames):
    out.write(frame)

out.release()

print("Reverse video created successfully.")
```

IDLE Shell 3.14.3

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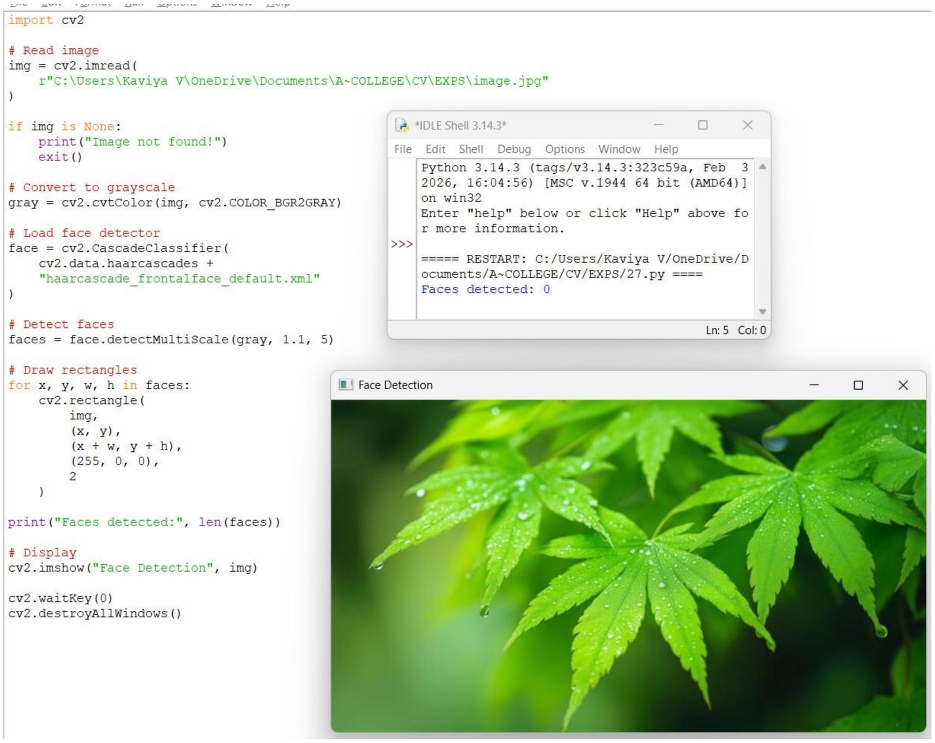
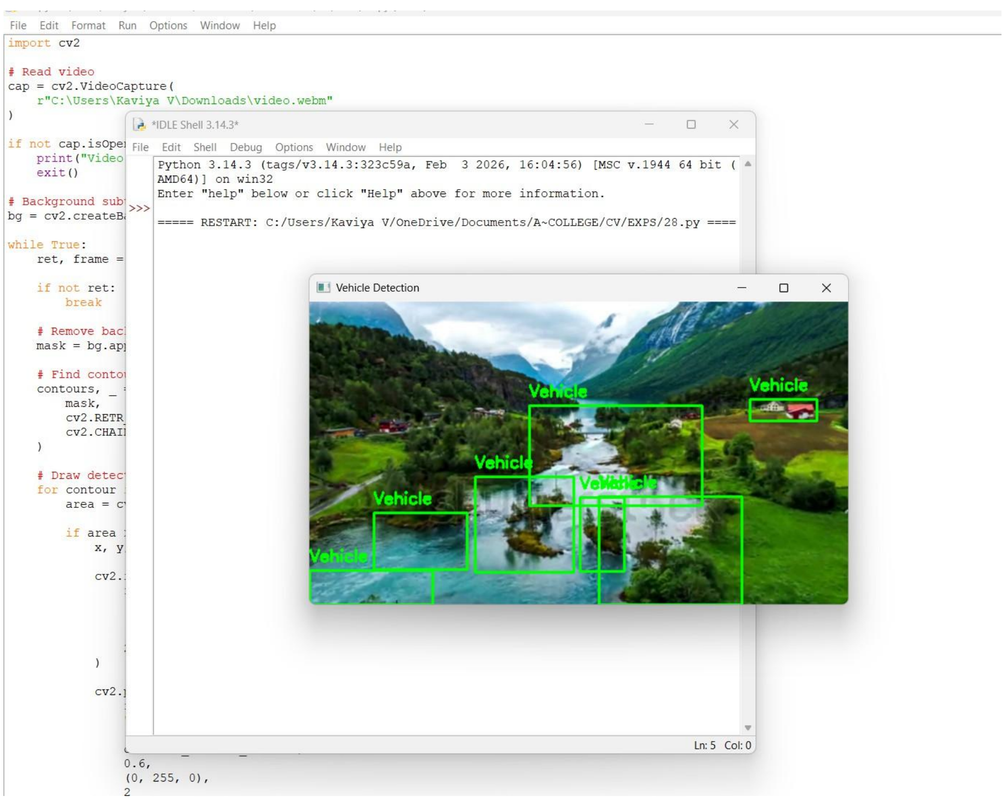
```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Kaviya V/OneDrive/Documents/A~COLLEGE/CV/EXPS/26.py ====
Reverse video created successfully.
>>>
```

Ln: 6 Col: 0

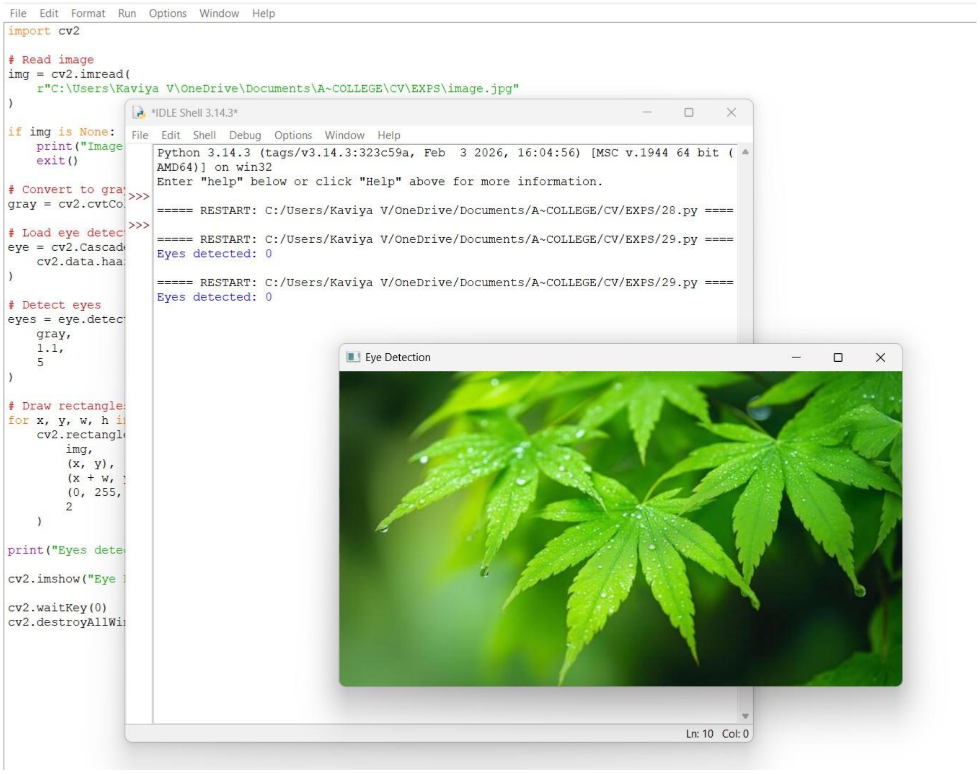
27. Implement a face detection algorithm using Open CV to detect and locate human faces in the images.




28. Implement a vehicle detection algorithm using Open CV to detect and locate vehicles in each frame of the video.



29. Implement an Eye detection algorithm using Open CV to detect and locate human eyes in the images.




30. Implement a Smile detection algorithm using Open CV to detect and locate human smile in the images.